# MPG No. 36.1.144 $\bar{3}m$ (-31m setting) [ Type I, trigonal ] [Q tensor]

* Rank 0 tensor.

$$[Q]$$

$$Q = \mathbb{Q}_0^{(1)}$$

* Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} Q_{xx} & 0 & 0 \\ 0 & Q_{xx} & 0 \\ 0 & 0 & Q_{zz} \end{bmatrix}$$

$$Q_{xx} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(1)}$$
$$Q_{zz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_u^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & 0 & Q_{xxzx} & 0 \\ Q_{xxyy} & Q_{xxxx} & Q_{xxzz} & 0 & -Q_{xxzx} & 0 \\ Q_{xxzz} & Q_{xxzz} & Q_{zzzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & Q_{yzyz} & 0 & -Q_{xxzx} \\ Q_{xxzx} & -Q_{xxzx} & 0 & 0 & Q_{yzyz} & 0 \\ 0 & 0 & 0 & -Q_{xxzx} & 0 & \frac{Q_{xxxx}}{2} - \frac{Q_{xxyy}}{2} \end{bmatrix}$$

$$Q_{xxxx} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 3\mathbb{Q}_{40}^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)}$$
$$Q_{xxyy} = \mathbb{Q}_0^{(1)} + \mathbb{Q}_{40}^{(1)} - 2\mathbb{Q}_u^{(1)}$$
$$Q_{xxzz} = \mathbb{Q}_0^{(1)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(1)}$$
$$Q_{xxzx} = \mathbb{Q}_{4c}^{(1)}$$
$$Q_{zzzz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 8\mathbb{Q}_{40}^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)}$$
$$Q_{yzyz} = \mathbb{Q}_0^{(2)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & Q_{xxzz} & 0 & Q_{xxzx} & 0 \\ 0 & 0 & Q_{xxzz} & 0 & -Q_{xxzx} & 0 \\ -Q_{xxzz} & -Q_{xxzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{xxzx} \\ -Q_{xxzx} & Q_{xxzx} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -Q_{xxzx} & 0 & 0 \end{bmatrix}$$

$$Q_{xxzz} = 3\mathbb{Q}_u^{(3)}$$
$$Q_{xxzx} = -2\mathbb{G}_{3s}^{(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} Q_{yzyz} & 0 & 0 \\ 0 & Q_{yzyz} & 0 \\ 0 & 0 & Q_{xyxy} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)}$$
$$Q_{xyxy} = \mathbb{Q}_0^{(3)} + 4\mathbb{Q}_u^{(4)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & Q_{xxzx} & 0 \\ 0 & -Q_{xxzx} & 0 \\ 0 & 0 & 0 \\ Q_{yzyz} & 0 & 0 \\ 0 & -Q_{yzyz} & 0 \\ Q_{xxzx} & 0 & 0 \end{bmatrix}$$

$$Q_{xxzx} = \mathbb{G}_{3s}^{(2)}$$
$$Q_{yzyz} = -3\mathbb{Q}_{u}^{(5)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & Q_{yzyz} & 0 & Q_{yzxy} \\ Q_{yzxy} & -Q_{yzxy} & 0 & 0 & -Q_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$Q_{yzyz} = -3\mathbb{Q}_{u}^{(6)}$$
$$Q_{yzxy} = \mathbb{G}_{3s}^{(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & 0 & Q_{xxzx} & 0 & 0 & Q_{xxxz} & 0 \\ Q_{xxyy} & Q_{xxxx} & Q_{xxzz} & 0 & -Q_{xxzx} & 0 & 0 & -Q_{xxxz} & 0 \\ Q_{zzxx} & Q_{zzxx} & Q_{zzzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & Q_{yzyz} & 0 & Q_{yzxy} & Q_{yzzy} & 0 & Q_{yzxy} \\ -Q_{yzxy} & Q_{yzxy} & 0 & 0 & Q_{yzzy} & 0 & 0 & Q_{yzyz} & 0 \\ 0 & 0 & 0 & -Q_{xxxz} & 0 & \frac{Q_{xxxx}}{2} - \frac{Q_{xxyy}}{2} & -Q_{xxzx} & 0 & \frac{Q_{xxxx}}{2} - \frac{Q_{xxyy}}{2} \end{bmatrix}$$

$$Q_{xxxx} = \mathbb{Q}_{0}^{(1)} + 2\mathbb{Q}_{0}^{(2)} + 3\mathbb{Q}_{40}^{(1)} - 2\mathbb{Q}_{u}^{(1)} - 4\mathbb{Q}_{u}^{(2)}$$
$$Q_{xxyy} = \mathbb{Q}_{0}^{(1)} + \mathbb{Q}_{40}^{(1)} - 2\mathbb{Q}_{u}^{(1)}$$
$$Q_{xxzz} = \mathbb{Q}_{0}^{(1)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_{u}^{(1)} + 3\mathbb{Q}_{u}^{(3)}$$
$$Q_{xxzx} = -2\mathbb{G}_{3s}^{(1)} + \mathbb{G}_{3s}^{(2)} + \mathbb{Q}_{4c}^{(1)}$$
$$Q_{xxxz} = -2\mathbb{G}_{3s}^{(1)} - \mathbb{G}_{3s}^{(2)} + \mathbb{Q}_{4c}^{(1)}$$
$$Q_{zzxx} = \mathbb{Q}_{0}^{(1)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_{u}^{(1)} - 3\mathbb{Q}_{u}^{(3)}$$
$$Q_{zzzz} = \mathbb{Q}_{0}^{(1)} + 2\mathbb{Q}_{0}^{(2)} + 8\mathbb{Q}_{40}^{(1)} + 4\mathbb{Q}_{u}^{(1)} + 8\mathbb{Q}_{u}^{(2)}$$
$$Q_{yzyz} = \mathbb{Q}_{0}^{(2)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_{u}^{(2)} - 3\mathbb{Q}_{u}^{(5)}$$
$$Q_{yzxy} = -2\mathbb{G}_{3s}^{(1)} - \mathbb{Q}_{4c}^{(1)}$$
$$Q_{yzzy} = \mathbb{Q}_{0}^{(2)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_{u}^{(2)} + 3\mathbb{Q}_{u}^{(5)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & Q_{yzyz} & 0 & Q_{yzxy} & Q_{yzzy} & 0 & Q_{yzxy} \\ Q_{yzxy} & -Q_{yzxy} & 0 & 0 & -Q_{yzzy} & 0 & 0 & -Q_{yzyz} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{xyxy} & 0 & 0 & -Q_{xyxy} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{Q}_{0}^{(3)} - 2\mathbb{Q}_{u}^{(4)} - 3\mathbb{Q}_{u}^{(6)}$$
$$Q_{yzxy} = \mathbb{G}_{3s}^{(3)}$$
$$Q_{yzzy} = -\mathbb{Q}_{0}^{(3)} + 2\mathbb{Q}_{u}^{(4)} - 3\mathbb{Q}_{u}^{(6)}$$
$$Q_{xyxy} = \mathbb{Q}_{0}^{(3)} + 4\mathbb{Q}_{u}^{(4)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} Q_{xxxx} & 0 & Q_{xxxz} \\ 0 & Q_{xxxx} & 0 \\ 0 & 0 & Q_{zzzz} \\ -Q_{xxxz} & 0 & Q_{yyzz} \\ Q_{zzxx} & 0 & 0 \\ 0 & Q_{xxyy} & 0 \\ 0 & Q_{yzzy} & 0 \\ Q_{xxxz} & 0 & -\frac{Q_{xxxx}}{2} + \frac{3Q_{xxyy}}{2} + Q_{yyzz} - Q_{yzzy} + Q_{zzxx} \\ \frac{Q_{xxxx}}{2} - \frac{Q_{xxyy}}{2} & 0 & -Q_{xxxz} \\ 0 & -Q_{xxxz} & 0 \end{bmatrix}$$

$$Q_{xxxx} = 3\mathbb{Q}_0^{(1)} + 3\mathbb{Q}_{40}^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)}$$
$$Q_{xxxz} = \mathbb{Q}_{4c}^{(1)}$$
$$Q_{zzzz} = 3\mathbb{Q}_0^{(1)} + 8\mathbb{Q}_{40}^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)}$$
$$Q_{yyzz} = \mathbb{Q}_0^{(1)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)}$$
$$Q_{zzxx} = \mathbb{Q}_0^{(1)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)}$$
$$Q_{xxyy} = \mathbb{Q}_0^{(1)} + \mathbb{Q}_{40}^{(1)} - 2\mathbb{Q}_u^{(1)}$$
$$Q_{yzzy} = \mathbb{Q}_0^{(1)} - 4\mathbb{Q}_{40}^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)}$$